

Integrating the Android app with Firebase

Firebase provides Java APIs for the Android platform to use its cloud services. To use these APIs, one needs to include Firebase libraries in the app's build scripts to point to the class path and compile-time dependencies. The three steps described below explain how to set up the necessary environment in Android Studio.

1. Since we are using an external library here, the *build.gradle* at project level (which should be in the root directory of your project) needs to be told about the *classpath* for Google services by mentioning it as a dependency in the buildscript as shown below:

```
classpath 'com.google.gms:google-services:3.0.0'
```

The buildscript block of *build.gradle* at the project level should now look like what's shown below, with the above addition:

```
buildscript {
    repositories {
        jcenter()
    }
    dependencies {
        classpath 'com.android.tools.build:gradle:1.5.0'
        classpath 'com.google.gms:google-services:3.0.0'
    }
}
```

2. The module's buildscript (*build.gradle* in the app folder) needs to add the Firebase library as a dependency:

```
compile 'com.google.firebase:firebase-database:9.2.1'
```

This library is for the real-time database's features. The dependencies block of the module's *build.gradle* will now look like what's shown below, with this addition:

```
dependencies {
    compile fileTree(dir: 'libs', include: ['*.jar'])
    testCompile 'junit:junit:4.12'
    compile 'com.android.support:appcompat-v7:23.2.1'
    compile 'com.android.support:design:23.2.1'
    compile 'com.google.firebase:firebase-database:9.2.1'
}
```

3. Finally, make sure that *AndroidManifest.xml* has permissions set for Internet access, as the app needs to contact the Firebase server for pushing and pulling real-time data.

All these steps are done on the SDK side. Now, you may have to register the app with a Firebase account. This can be done with the following steps:

1. Log in to your account from <https://firebase.google.com/>

using your Google credentials. The site has support for both Firebase's legacy and its latest consoles. This whole article is based on the latest styled consoles. The consoles here are used to access your account to check the real-



Figure 1: Post login screen of the Firebase account

- time database to which the app will be writing.
2. You might see the post login screen as shown in Figure 1. Click on 'Create New Project' to start with the Firebase project.
3. A new project gets created under your account, and you can view and use different Firebase services using the

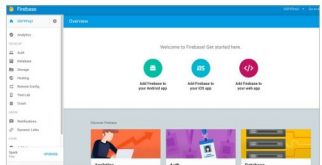


Figure 2: Project's dashboard

project's dashboard as shown in Figure 2. These different services are seen on the left side column of the dashboard as shown in this figure. This article deals with the database service for dealing with real-time data.

4. Now the mechanism in which the app communicates with your Firebase account during its run time needs to be set up. For this, we need to register the Android app

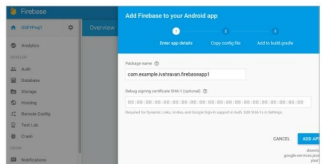


Figure 3: Adding Firebase to the Android project